

10ALYTICS HEALTHCARE DATA ANALYTICS CAPSTONE

VitalLink Telehealth Solutions

VitaStat Analytics — Remote Patient Monitoring Optimization

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Role: Lead Healthcare Data Analyst (capstone simulation)

Tool: Tableau — 2-page interactive dashboard

Dataset: Vital_Alert_Fact (10,500 alerts) joined to Dim_Patient, Dim_Clinician, Dim_Device

CONTEXT FROM THE PROJECT BRIEF

VitalLink is experiencing an alert processing bottleneck that is affecting clinician response times, operational efficiency, and patient safety.

Alert Overload ("Alarm Fatigue")

Critical and non-critical vital sign alerts are mixed into a single feed, which risks desensitising nursing staff to genuinely urgent alerts.

SLA Blind Spots

Management cannot currently see which regional clinical hubs are missing the target of reviewing high-priority alerts within 2 hours.

Device Status Disconnect

There is no visibility on whether older device models or poor battery status are contributing to a surge in warning-level alerts.

My role, as Lead Healthcare Data Analyst

- Normalise the raw telemetry records into a high-performance star schema (Vital_Alert_Fact joined to Dim_Patient, Dim_Clinician, Dim_Device).
- Write calculated fields to track clinician response times, SLA compliance margins, and device fleet trends.
- Build a 2-page interactive Tableau dashboard covering Patient Alert Profiles and Device Health / Clinician SLA performance.

WHAT THE ANALYSIS IS BUILT ON

A star schema joining one fact table of telemetry alerts to three dimension tables, connected on ClinicianID, DeviceID and PatientID.

10,500
ALERTS (FACT TABLE)

4,417
DISTINCT PATIENTS

12
CLINICIANS / 4 HUBS

40
DEVICES / 4 MODELS

Key calculated fields

- SLA Status — Response Time Hrs $\leq$ 2 = "Within SLA", else "Breached"
- SLA Breaches / SLA Breach % — COUNTD of breached alerts
- SLA Compliance % — FIXED per clinician
- Age Group — banded from Patient Age (45–54, 55–64, 65–74, 75+)
- Battery Band — banded from Battery Status (0–25% ... 75–100%)

How the two dashboard pages are organised

Page 1 — Patient Alerts Profile: alert volume, severity mix, vital type, age group & condition, patient city.

Page 2 — Device Health & Clinician Efficiency: hub response time vs. the 2-hour SLA, SLA breach volume, clinician workload & compliance, battery and connectivity patterns behind warning alerts.

Patient Alerts Profile

Who is generating alerts, how severe are they, and where are they concentrated? The next four slides walk through this page one business question at a time, using the figures actually returned by the workbook.

VL VitalLink Telehealth — Patient Alerts Profile

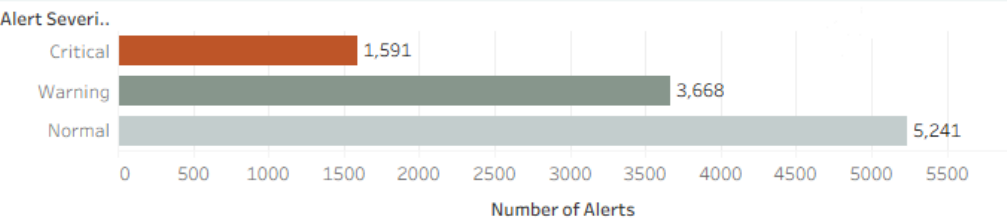
Total Alerts &
Critical %

10,500
15.2% Critical

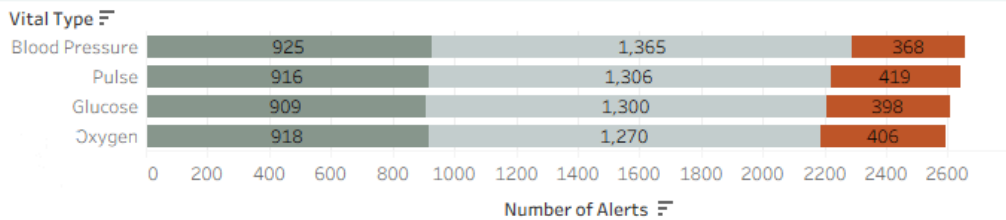
SLA Breaches

8,425
80.2% Breached

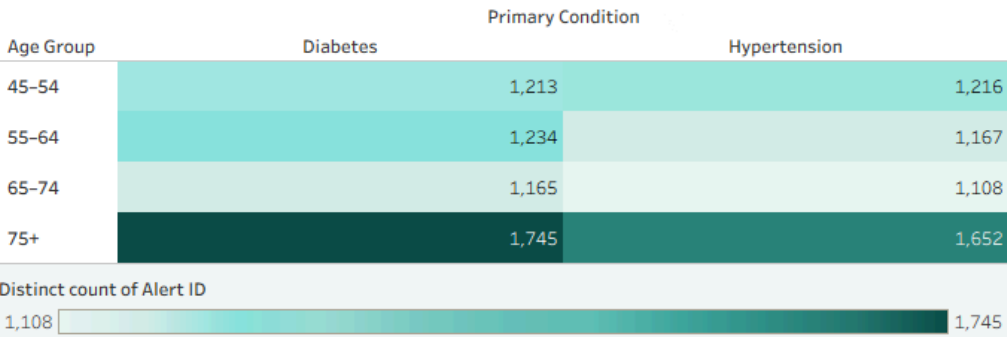
Alert Volume by Severity



Alert by Vital Type

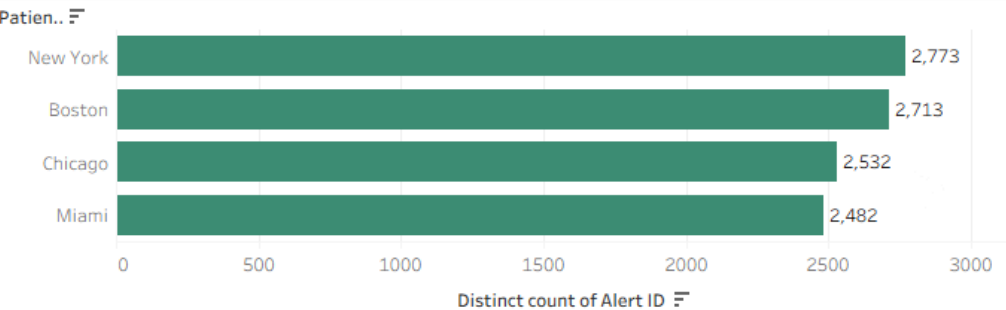


Alerts by Age Group & Condition



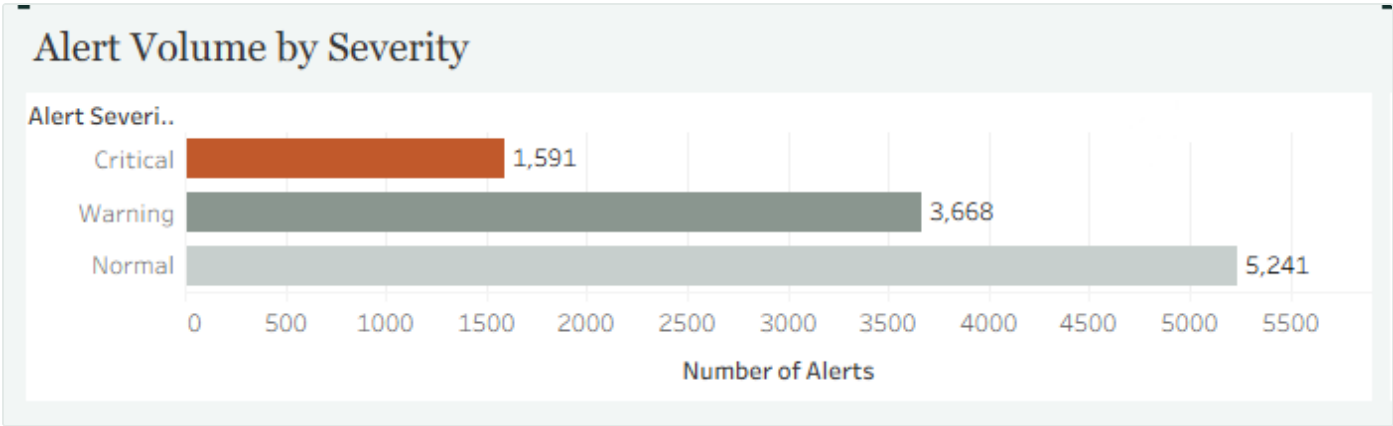
Alert Severity: Critical (orange), Warning (dark green), Normal (light green)

Alerts by City



BUSINESS QUESTION

What is the total volume of alerts generated, and what percentage fall under the Critical category?



TOTAL ALERTS
10,500

CRITICAL ALERT %
15.2%
1,591 of 10,500 alerts

WHAT THE DATA SHOWS

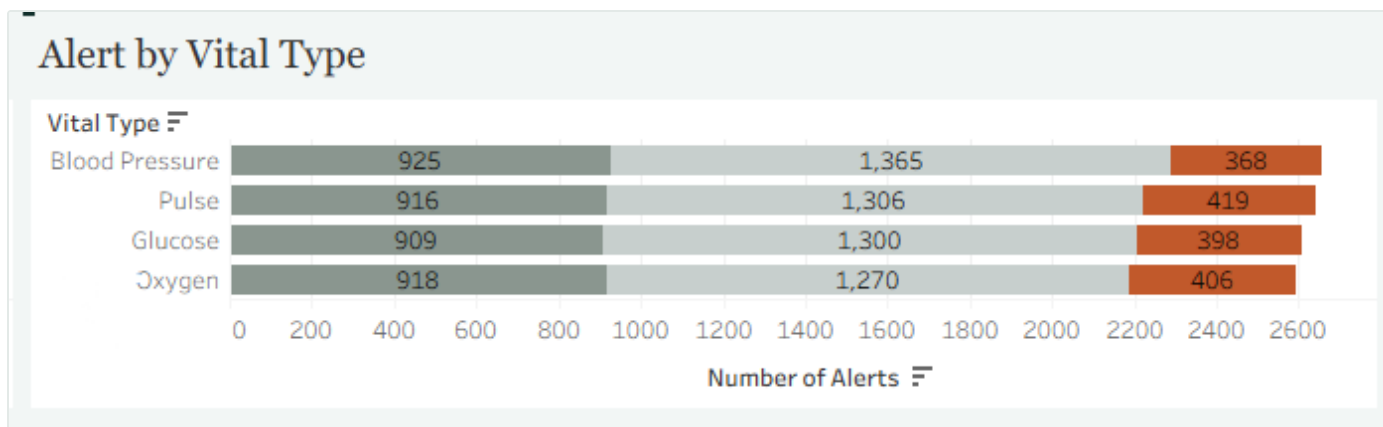
Alerts split 3 ways by severity: 5,241 Normal, 3,668 Warning, 1,591 Critical. Critical alerts are the minority (15.2%) of total volume — the majority of the queue is Normal/Warning.

WHY IT MATTERS

Because Critical alerts sit inside a much larger stream of lower-priority ones, this is exactly the "alarm fatigue" risk named in the brief: without severity-based triage, urgent alerts can be missed among routine ones.

BUSINESS QUESTION

Which vital sign type (Blood Pressure, Glucose, Pulse, or Oxygen) triggers the highest number of system alerts?



WHAT WE'RE LOOKING AT

Total alert volume broken down by the vital sign that triggered it, further split by severity (Critical / Warning / Normal) within each type.

WHAT THE DATA SHOWS

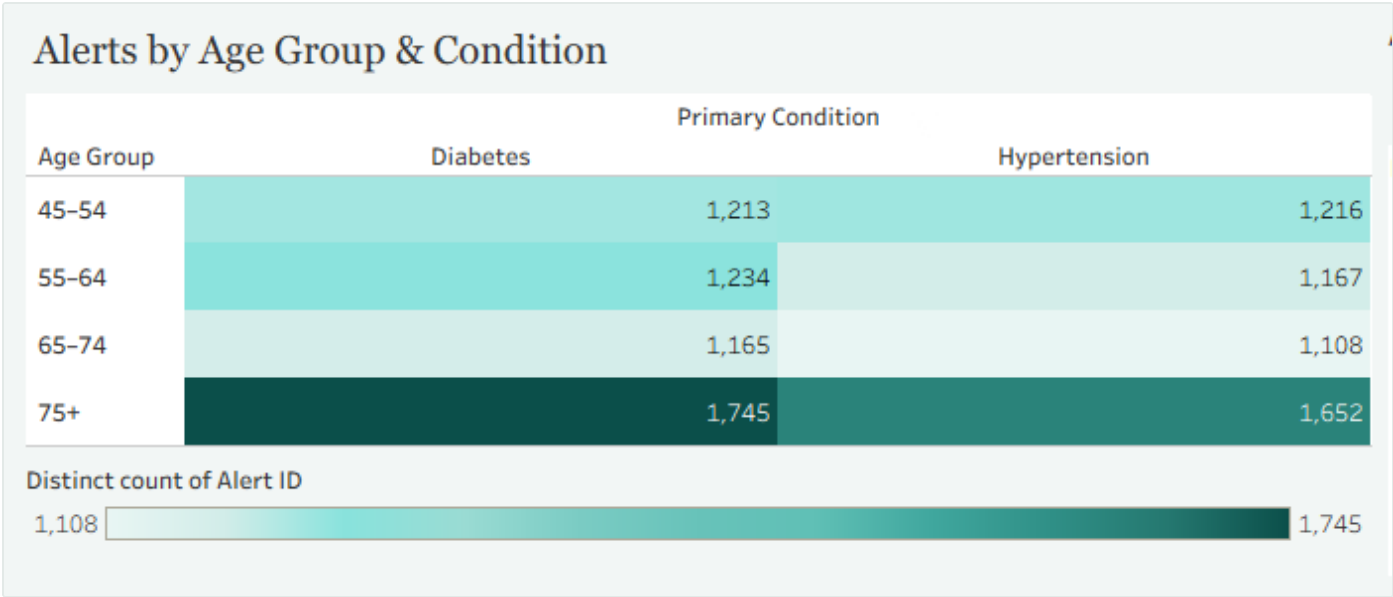
Volume is almost perfectly even across all four vital types: Blood Pressure 2,658, Pulse 2,641, Glucose 2,607, Oxygen 2,594 total alerts. Critical counts are similarly close — 368 to 419 per type.

KEY INSIGHT

No single vital sign dominates alert volume. This suggests the alert overload is a general monitoring-threshold issue across all tracked vitals, not a fault specific to one type of sensor or reading.

BUSINESS QUESTION

How do alerts distribute across different patient age groups and primary conditions (Diabetes vs. Hypertension)?



WHAT WE'RE LOOKING AT

A heatmap of alert volume by patient Age Group (rows) and Primary Condition — Diabetes or Hypertension (columns). Darker cells = more alerts.

WHAT THE DATA SHOWS

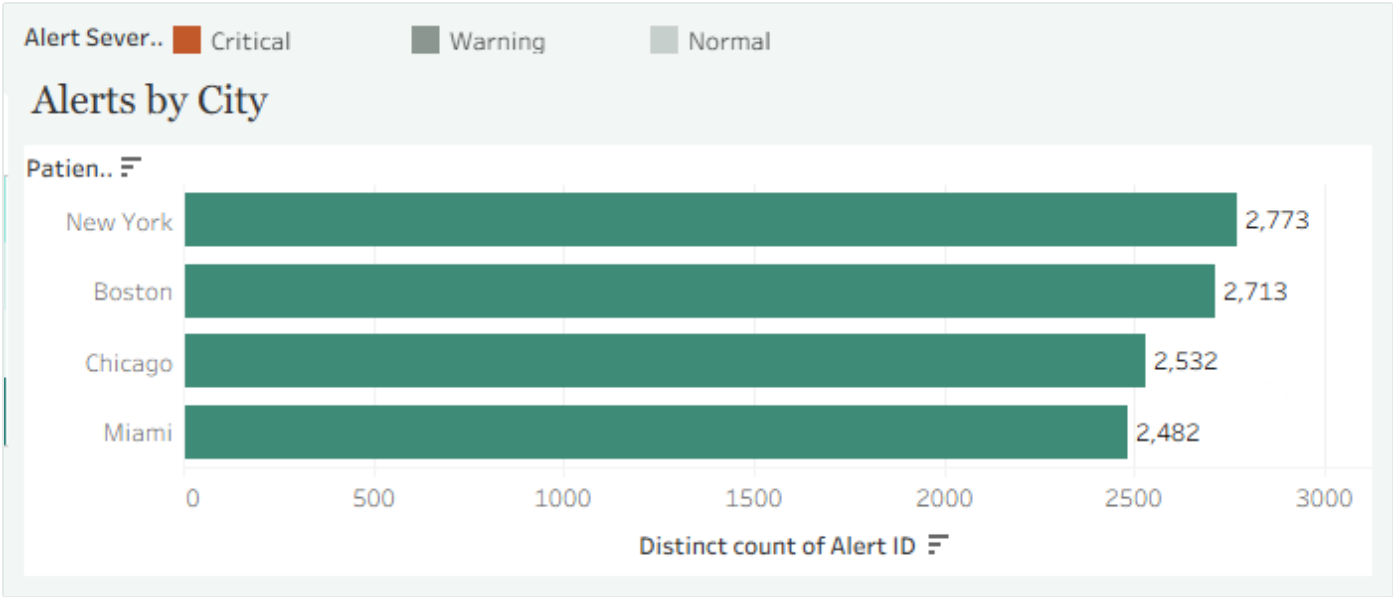
The 75+ age group generates the most alerts by far (3,397 total: 1,745 Diabetes, 1,652 Hypertension) — noticeably higher than the other three age bands, which each sit close together at 2,270–2,430.

KEY INSIGHT

Within every age group, Diabetes and Hypertension generate a near-even split of alerts. The clearest pattern is by age, not condition: alert volume rises with patient age, concentrated most heavily in the oldest cohort.

BUSINESS QUESTION

Which patient cities have the highest alert volumes, and how does Critical-alert share compare across cities?



WHAT WE'RE LOOKING AT

Total alert volume by patient city. (Critical-alert share by city, computed from the same fact table, is reported alongside for this specific question.)

WHAT THE DATA SHOWS

New York (2,773), Boston (2,713), Chicago (2,532) and Miami (2,482) are all within a narrow band of one another. Critical-alert share by city is similarly close: Chicago 16.3%, New York 15.1%, Miami 15.1%, Boston 14.2%.

KEY INSIGHT

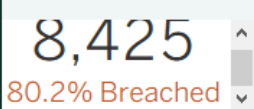
No single city is a disproportionate driver of alert volume or of Critical-severity alerts specifically — Chicago runs slightly highest on Critical share, but the gap to the other three cities is modest rather than dramatic.

Device Health & Clinician Efficiency

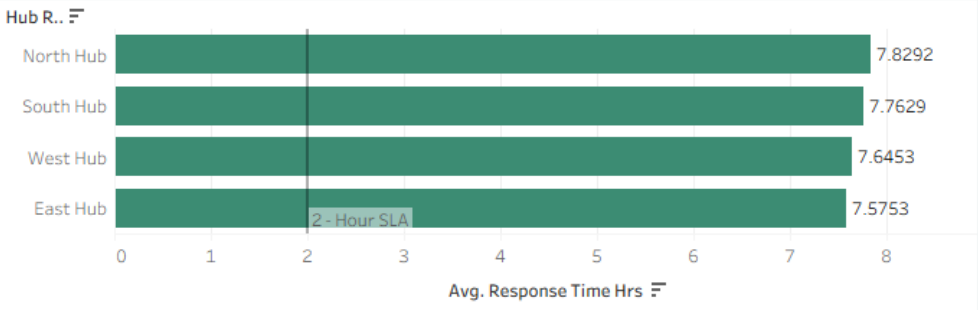
Are clinicians meeting the 2-hour SLA, is workload balanced across the team, and do device battery or connectivity patterns explain the volume of warning alerts? The next four slides answer each question with the actual dashboard figures.

VL VitalLink Telehealth — Device Health & Clinician Efficiency

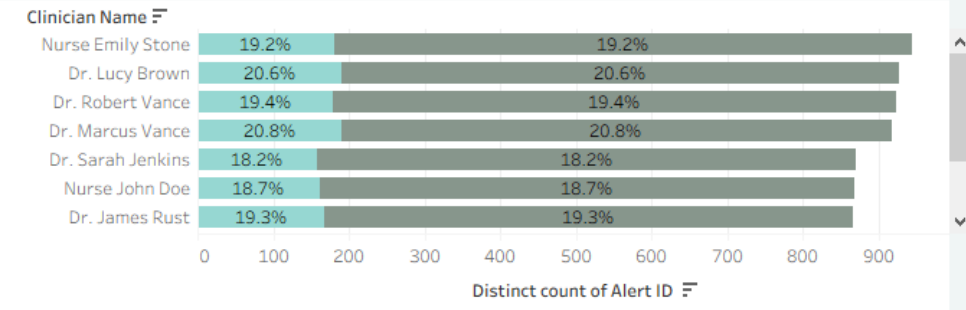
SLA Breaches



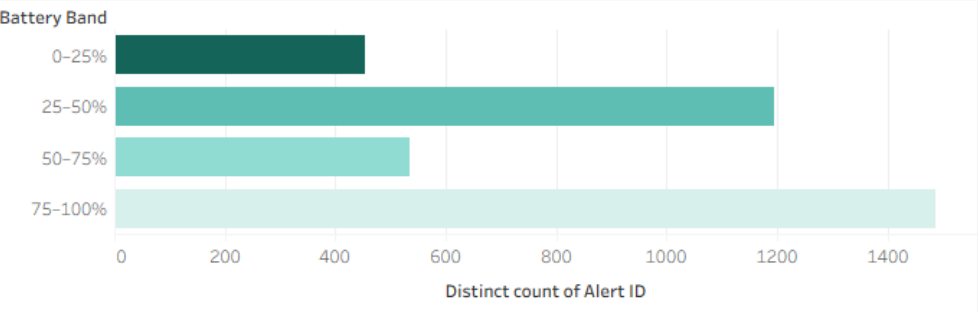
Avg. Response Time by Hub



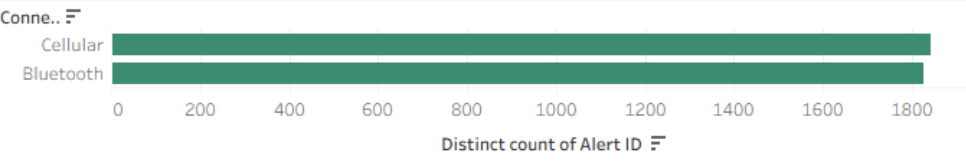
Clinician Workload & Compliance



Warning Alerts by Battery Status

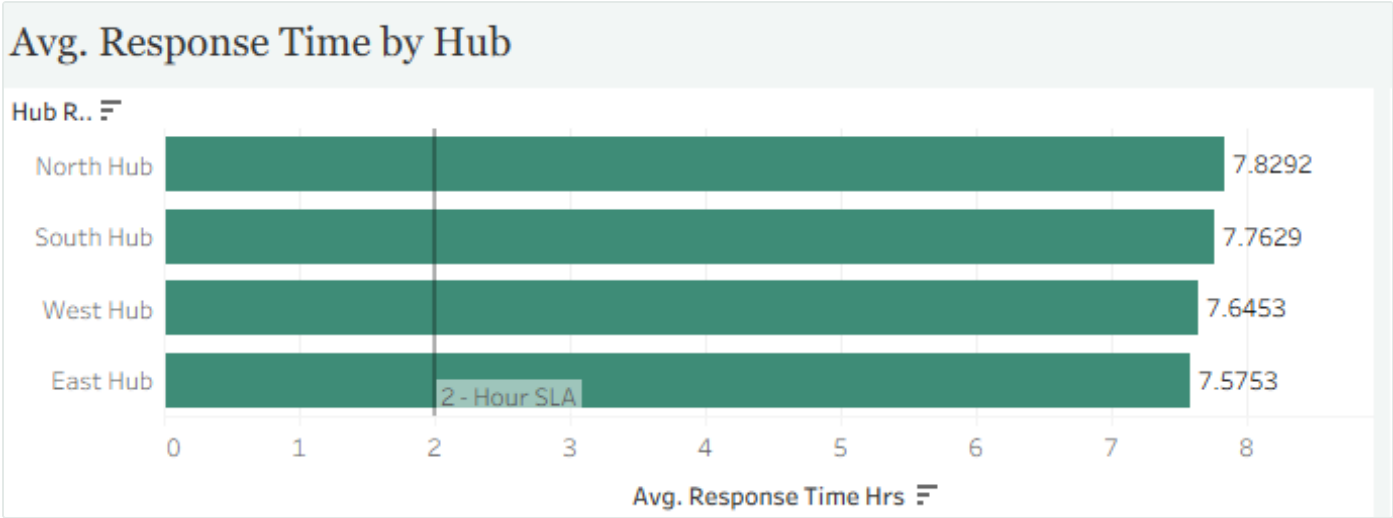


Warning Alerts by Connectivity



BUSINESS QUESTION

What is the average clinician response time across our 4 primary operating regional hubs?



WHAT WE'RE LOOKING AT

Average clinician response time (hours) by regional hub, plotted against a 2-hour SLA reference line.

WHAT THE DATA SHOWS

All four hubs average 7.6–7.8 hours to respond — North Hub 7.83, South Hub 7.76, West Hub 7.65, East Hub 7.58. Every hub sits nearly 4x above the 2-hour target, and the spread between the best and worst hub is only 0.25 hours.

WHY IT MATTERS

Because all four hubs miss the SLA by a similar margin, this points to a system-wide process issue rather than one under-performing region.

BUSINESS QUESTION

How many alerts breached the mandatory 2-hour clinical response deadline?

SLA BREACHES

8,425

of 10,500 total alerts

BREACH RATE

80.2%

AVG. RESPONSE TIME BY SEVERITY, VS. THE 2-HOUR TARGET



WHAT THE DATA SHOWS

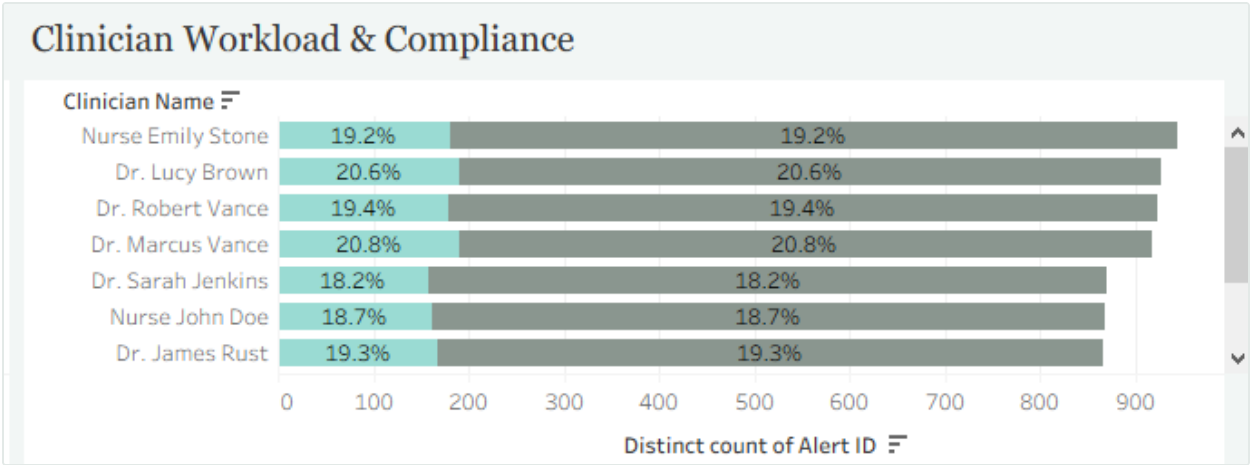
80.2% of all alerts took longer than 2 hours to review. But average response time by severity tells a more precise story: Critical alerts average 1.82 hours — inside the SLA — while Warning alerts average 3.28 hours and Normal alerts average 12.58 hours.

KEY INSIGHT

The high overall breach rate is driven almost entirely by Normal- and Warning-severity alerts sitting unreviewed for much longer than Critical ones — Critical alerts have an average response time within the 2-hour SLA, while Warning and Normal alerts exceed it.

BUSINESS QUESTION

Which individual clinicians are managing the highest volume of alerts, and what is their compliance rate?



Clinician	Alerts	Compliance
Nurse Emily Stone	944	19.2%
Dr. Lucy Brown	928	20.6%
Dr. Robert Vance	924	19.4%
Dr. Marcus Vance	917	20.8%
Dr. Sarah Jenkins	870	18.2%
Nurse John Doe	868	18.7%
Dr. James Rust	866	19.3%
Dr. Ellen Choi	858	21.1%
Dr. Alan Davies	847	19.7%
Dr. Linda Lawson	845	21.1%
Nurse Charlotte Bowen	835	19.5%
Dr. Aris Patel	798	19.7%

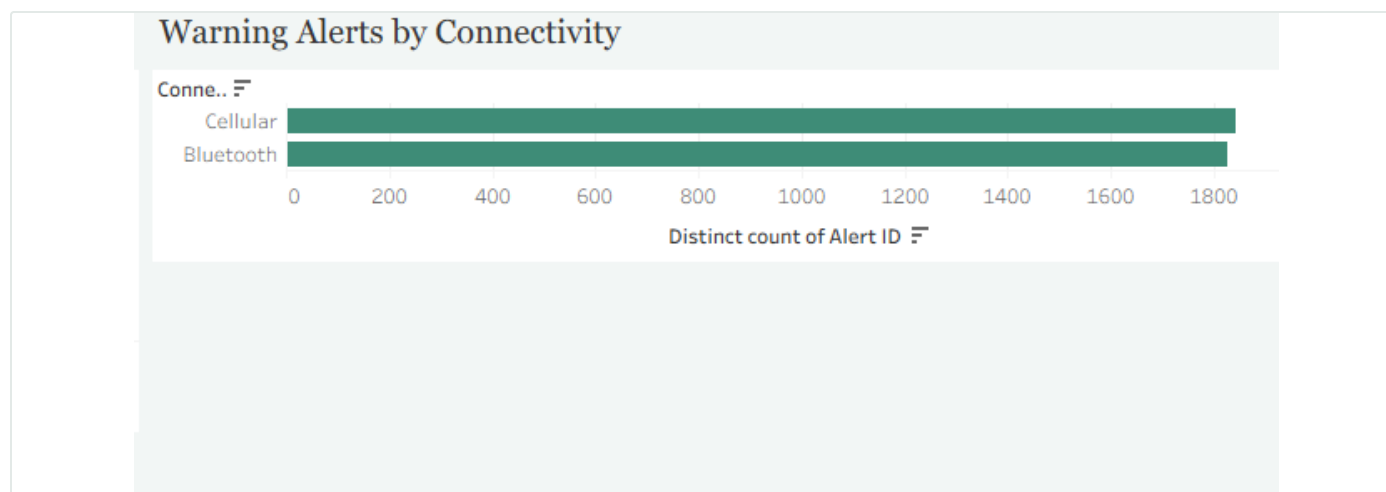
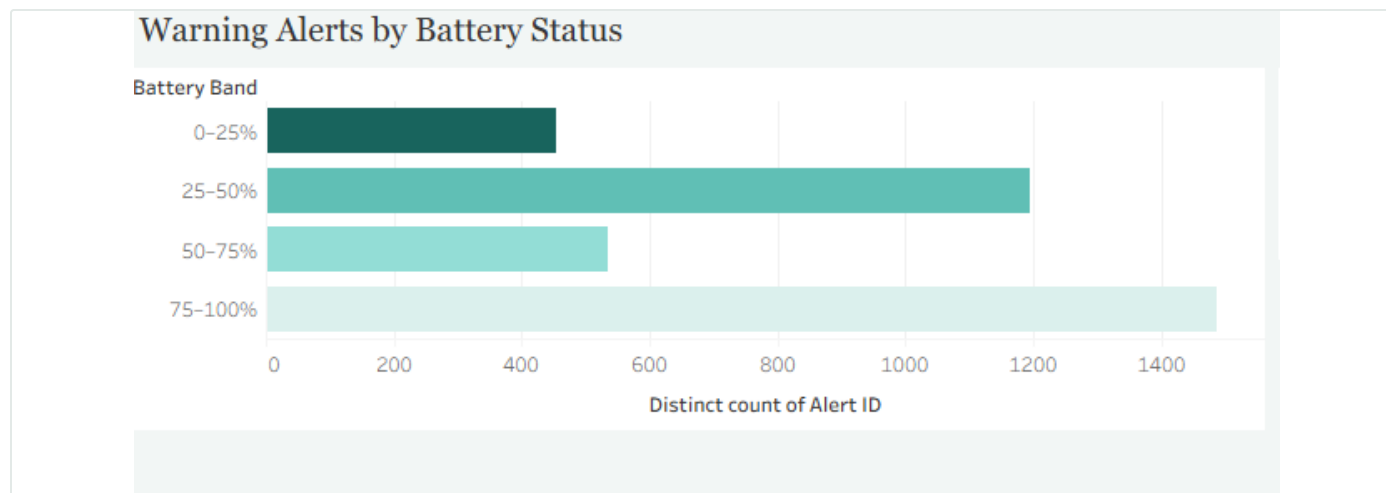
Full clinician list (all 12), computed from the same fact table shown in the Tableau view at left.

KEY INSIGHT

Workload is fairly balanced (798–944 alerts per clinician) and compliance is uniformly low across the whole team (18.2%–21.1%). No individual clinician is an outlier in either direction — this again points to a process-level SLA issue rather than an individual performance issue.

BUSINESS QUESTION

Is there a visible correlation between low device battery, connectivity type, and the volume of warning alerts?



WHAT THE DATA SHOWS

Warning alerts are highest at 75–100% battery (1,487) and lowest at 0–25% battery (453) — the opposite of what a "low battery causes warnings" hypothesis would predict. By connectivity, volume is almost identical: Cellular 1,843 vs. Bluetooth 1,825, and Critical-alert rate is nearly the same for both (15.0% vs. 15.3%).

KEY INSIGHT

In this dataset, low battery and connectivity type do not show a meaningful correlation with warning-alert volume or Critical-alert rate. The data does not support device battery or connection type as the primary driver of the warning-alert surge.

ACROSS BOTH DASHBOARD PAGES

A summary of what the data shows across every dimension analysed — alert profile, patient characteristics, hub & clinician performance, and device patterns.

ALERT MIX

Critical alerts are 15.2% of volume (1,591 of 10,500); Normal alerts are the majority (5,241), consistent with an alarm-fatigue risk from mixing severities in one feed.

VITAL TYPE

Alert volume is nearly identical across Blood Pressure, Pulse, Glucose and Oxygen (2,594–2,658 each) — no single vital sign dominates.

AGE & CONDITION

The 75+ age group drives noticeably more alerts (3,397) than any other age band; Diabetes and Hypertension split evenly within every age group.

CITY

Alert volume and Critical-alert share are both fairly even across New York, Boston, Chicago and Miami; Chicago is marginally highest on Critical share (16.3%).

HUB RESPONSE TIME

All 4 hubs average 7.6–7.8 hours to respond, well above the 2-hour target, with only a 0.25-hour spread between hubs — a systemic issue, not one hub.

SLA BREACHES

80.2% of alerts breach the 2-hour SLA overall, but Critical alerts alone average 1.82 hours (within target); Normal alerts average 12.58 hours and are the main driver of the breach rate.

CLINICIAN WORKLOAD

Workload (798–944 alerts) and SLA compliance (18.2%–21.1%) are both consistent across all 12 clinicians — no individual outlier.

DEVICE PATTERNS

Warning-alert volume is highest at 75–100% battery, not low battery; Cellular vs. Bluetooth shows almost no difference in volume or Critical rate.

EVIDENCE-BASED, FOLLOWING FROM THE FINDINGS ABOVE

Each recommendation is tied to a specific finding from the dashboard — none are introduced without supporting evidence.

FINDING SLA breaches are concentrated in Normal/Warning alerts (avg. 12.58h / 3.28h), not Critical ones (avg. 1.82h).	IMPLICATION The current queue does not visibly separate urgency at the point of triage, so lower-priority alerts sit unreviewed for very long periods and inflate the headline breach rate.	RECOMMENDATION Consider a severity-based routing or escalation rule so Normal/Warning alerts are batched or reviewed on a longer, defined cycle, rather than competing unsorted with Critical alerts.
FINDING All 4 hubs breach the 2-hour SLA by a similar margin (7.58–7.83 hrs avg.), and all 12 clinicians show similar compliance (18.2%–21.1%).	IMPLICATION The consistent pattern across hubs and clinicians suggests that the issue should be investigated at a shared process/workflow level rather than treated solely as an individual or regional performance issue.	RECOMMENDATION Review the 2-hour target and current staffing/workflow at a process level across all hubs, rather than addressing it hub-by-hub or clinician-by-clinician.
FINDING Warning alerts are highest at 75–100% battery and connectivity type shows almost no difference in Critical-alert rate (15.0% vs 15.3%).	IMPLICATION The data does not support the working assumption that low battery or a specific connectivity type is driving the volume of warning alerts.	RECOMMENDATION Investigate other candidate causes of warning-alert volume (e.g. device model, calibration, or alert-threshold settings) rather than prioritising battery or connectivity fixes.
FINDING The 75+ patient age group generates substantially more alerts (3,397) than younger cohorts.	IMPLICATION Consider reviewing whether current alert thresholds and monitoring workflows are appropriate for the 75+ cohort, given its substantially higher alert volume.	RECOMMENDATION Consider reviewing whether current alert thresholds are age-adjusted, particularly for the 75+ cohort.

VL VitalLink Telehealth — Patient Alerts Profile

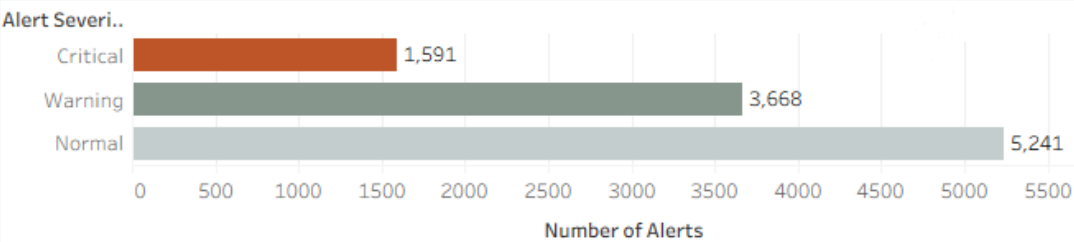
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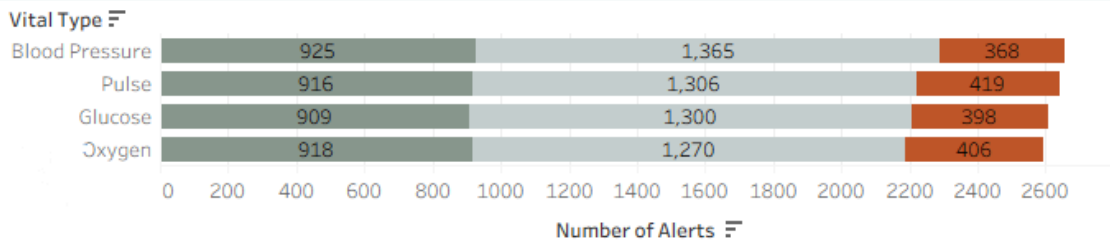
SLA Breaches

8,425
80.2% Breached

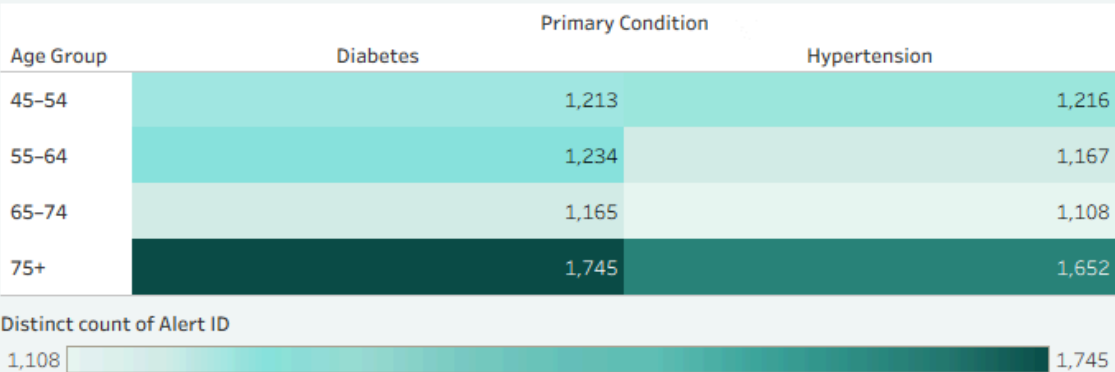
Alert Volume by Severity



Alert by Vital Type

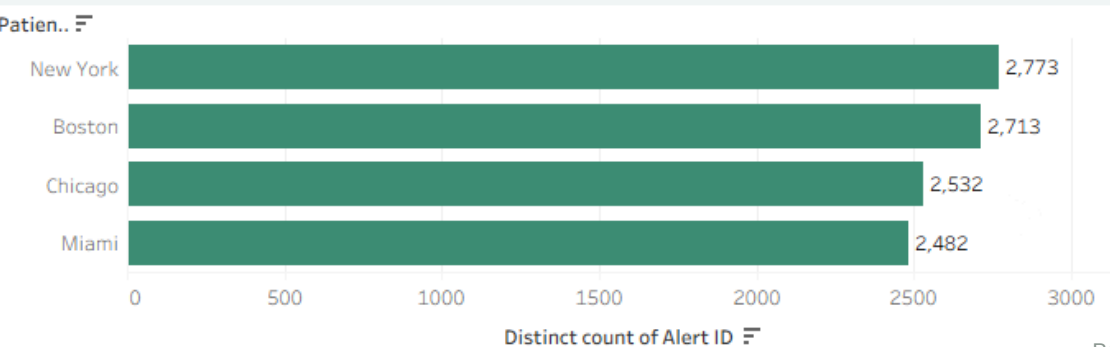


Alerts by Age Group & Condition



Alert Sever.. Critical Warning Normal

Alerts by City

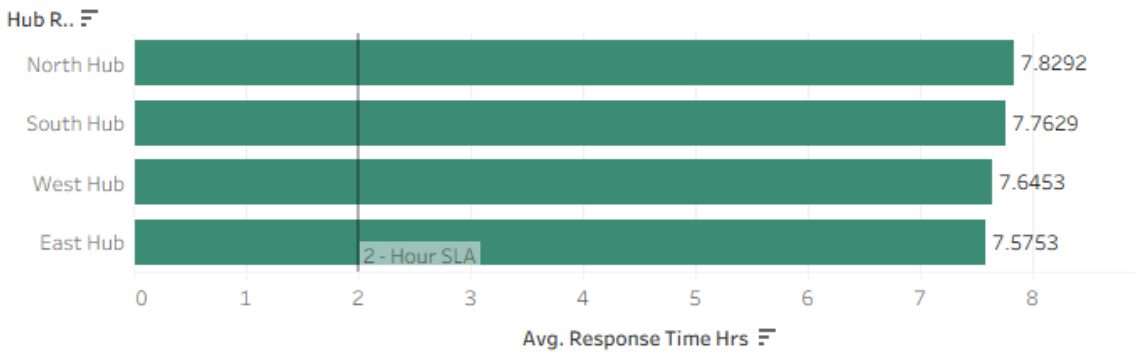


VL VitalLink Telehealth — Device Health & Clinician Efficiency

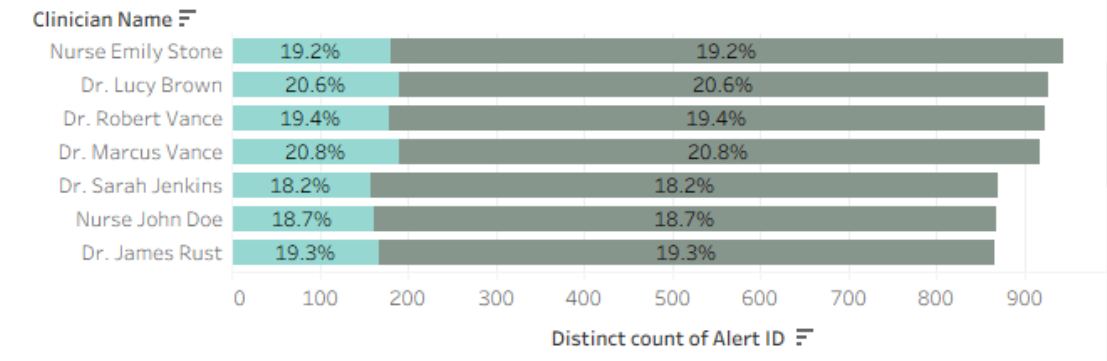
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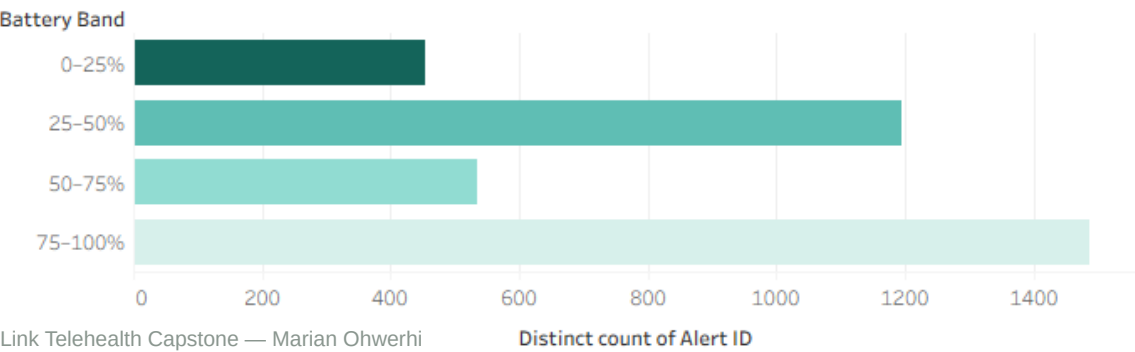
Avg. Response Time by Hub



Clinician Workload & Compliance



Warning Alerts by Battery Status



Warning Alerts by Connectivity

